

Space age

SpaceX's reusable rocket Falcon has had its share of disasters, one quite recent. Should we be worried?

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**Machine learning: Artificial Neural Networks**

Before we move on to what understanding emotion is and will mean, let's talk a bit about the techniques 'intelligent' machines currently use. To provide computers with intelligence, researchers look at a structure that has time and again proven itself to be a pinnacle of complexity and learning ability – the human brain. Nestled within our brains are a multitude of neurons, each of which interact and communicate in an intricate fashion to give us memory and processing abilities. But memory and processing components lie within every CPU box as well. What makes a human brain radically different from the world on a motherboard chip is that the neurons in the human brain network in a fashion that allows them not only to respond to stimuli, but also to continually learn from the input information. So, scientists replicated the working of a human brain on an algorithmic level and that led to the birth of an Artificial Neural Network (ANN). What an ANN essentially does is study a large set of input-output combinations to train itself, much like a child learns to identify an apple by seeing it a number of times. Once the network is trained, each input is processed and a new input is mapped to an output. Over time, because of feedback, and much like ourselves, the network corrects errors and gets better at dealing with inputs that weren't part of the original training set. This process can be broken down with a simple example. Any image is made of numbers, bytes representing values of color intensity for example. Now, an image of a red ball will have a certain input matrix of numbers. The fact that it is a ball can be represented with another unique marker matrix. The job of an ANN is to breakdown an image into raw data - an input function and after processing that information through its vast sieve of hidden layers, construct a corresponding output function. The output function will tell the ANN if the input corresponds to a ball or a cat. The same fundamental method is adopted by all deep learning machines. A chess

WILL ROBOTS EVER CRY?

Bots are cruel and heartless, following cold hard logic. But can they ever be programmed to understand emotion, feelings and other sentimental stuff? Let's find out...

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In 1837 Charles Babbage came up with the Analytical Engine, essentially the first machine design that laid claim to include all the elements of a modern computer. Back then Babbage's machine was designed to compute, crunch numbers and perform calculations. Today the landscape of modern computing has undergone a paradigm shift. Sure, we have computers that are fast and mean machines designed purely for number crunching, but now they can do so much more.

Traditionally the dividing line between a human and a computer has always been the ability of the former to process information to adapt and learn. But artificial intelligence, built on the bedrock of Turing's and Von Neumann's work, can now simulate human behaviour in not only absorbing, storing and even making sense of information. This ability to learn has allowed machines to be adept at image recognition, speech and text recognition and even games like Chess and Go. But a question that has been plaguing scientists and futurists alike for sometime is: Can machines understand emotion?